TimeR^4: Time-aware Retrieval-Augmented Large Language Models for Temporal Knowledge Graph Question Answering

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Abstract

Temporal Knowledge Graph Question Answering (TKGQA) aims to answer temporal questions using knowledge in Temporal Knowledge Graphs (TKGs). Previous works employ pre-trained TKG embeddings or graph neural networks to incorporate the knowledge of TKGs. However, these methods fail to fully understand the complex semantic information of time constraints. In contrast, Large Language Models (LLMs) have shown exceptional performance in knowledge graph reasoning, unifying both semantic understanding and structural reasoning. To further enhance LLMs' temporal reasoning ability, this paper aims to integrate temporal knowledge from TKGs into LLMs through a Time-aware Retrieve-Rewrite-Retrieve-Rerank framework, which we named **TimeR**^{4}. Specifically, to reduce temporal hallucination in LLMs, we propose a *retrieve-rewrite* module to rewrite questions using background knowledge stored in the TKGs, thereby acquiring explicit time constraints. Then, we implement a *retrieve-rerank* module aimed at retrieving semantically and temporally relevant facts from the TKGs and reranking according to the temporal constraints. To achieve this, we fine-tune a retriever using the contrastive time-aware learning framework. Our approach achieves great improvements, with relative gains of 47.8% and 22.5% on two datasets, underscoring its effectiveness in boosting the temporal reasoning abilities of LLMs. Our code is available at https://github.com/qianxinying/TimeR4 .

1 Introduction

Knowledge graph question answering (KGQA) aims to provide answers based on knowledge from knowledge graphs (Sun et al., 2019). However, many real-world questions include temporal constraints, such as "Who is the president of the United States after Obama?" To address this, some knowledge graphs store time-aware facts as quadruples *(subject, predicate, object, timestamp)*, which are known as temporal knowledge graphs (TKGs). Temporal knowledge graph question answering (TKGQA) focuses on obtaining answers using the knowledge in TKGs (Saxena et al., 2021).

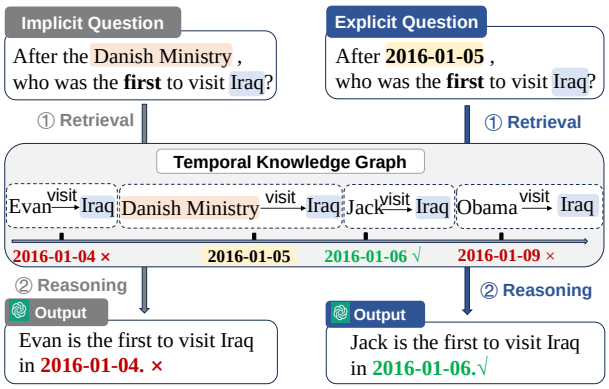


Figure 1: Examples of challenges in integrating temporal knowledge graphs with large language models.

Recent works (Saxena et al., 2021; Mavromatis et al., 2022) incorporate knowledge from TKGs by utilizing pre-trained TKG embeddings or graph neural networks (GNNs). However, these methods fail to fully understand the complex semantic information of time constraints in questions (Chen et al., 2023b). In contrast, Large language models (LLMs) have demonstrated exceptional performance in knowledge graph reasoning (Sun et al., 2024; Luo et al., 2024) and can unify semantic understanding and graph reasoning (Huang and Chang, 2023; Wei et al., 2023). To further enhance the temporal reasoning capabilities of LLMs, in this paper, we aim to integrate temporal knowledge into LLMs, thereby addressing complex and multi-granularity temporal questions. However, enhancing the temporal reasoning capabilities within LLMs remains several significant challenges:

(1) Hallucinated by implicit temporal ques-

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TimeR^4：面向时间感知的增强检索大型语言模型用于时间知识图谱问答

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摘要

时态知识图谱问答（TKGQA）旨在利用时态知识图谱（TKGs）中的知识回答时态问题。以往的工作使用预训练的TKG嵌入或图神经网络来整合TKGs的知识。然而，这些方法未能充分理解时间约束的复杂语义信息。相比之下，大型语言模型（LLMs）在知识图谱推理方面表现出色，实现了语义理解与结构推理的统一。为了进一步增强LLMs的时态推理能力，本文旨在通过一个时间感知的检索-重写-检索-重排序框架，将TKGs中的时态知识整合到LLMs中，我们称之为TimeR^4。具体来说，为了减少LLMs中的时态幻觉，我们提出了一个*retrieve-rewrite*模块，利用存储在TKGs中的背景知识重写问题，从而获得明确的时间约束。然后，我们实现了一个*retrieve-rerank*模块，旨在重新…

1 引言

知识图谱问答（KGQA）旨在基于知识图谱提供答案（Sun 等，2019）。然而，许多现实世界的问题包含时间约束，例如“谁是现任总统”

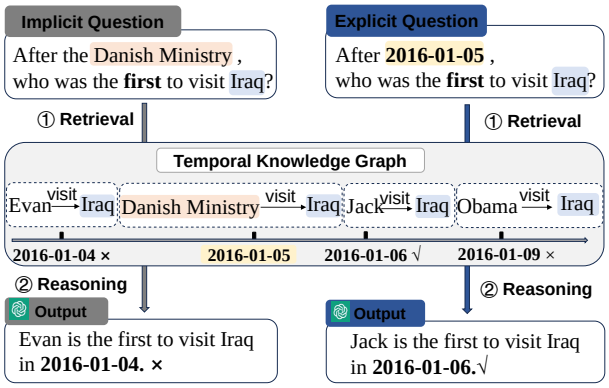


图1：将时序知识图谱与大型语言模型整合时的挑战示例。

奥巴马之后的各州？为了解决这个问题，一些知识图将时间相关的事实存储为四元组*(subject, predicate, object, timestamp)*，这些被称为时间知识图（TKGs）。时间知识图问答（TKGQA）侧重于使用 TKGs 中的知识来获取答案（Saxena 等，2021）。

近期的研究（Saxena 等，2021；Mavromatis 等，2022）通过利用预训练的时序知识图（TKG）嵌入或图神经网络（GNN）来整合来自 TKG 的知识。然而，这些方法未能充分理解问题中时间约束的复杂语义信息（Chen 等，2023b）。相比之下，大型语言模型（LLMs）在知识图推理方面表现出卓越的性能（Sun 等，2024；Luo 等，2024），并且能够统一语义理解和图推理（Huang 和 Chang，2023；Wei 等，2023）。为了进一步增强 LLM 的时间推理能力，本文旨在将时间知识整合到 LLM 中，从而解决复杂的、多粒度的时间问题。然而，在 LLM 中增强时间推理能力仍面临若干重大挑战：

(1) 由隐含的时间线索引起的幻觉

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tions. In the TKGQA task, there are many implicit questions, such as *"After the Danish Ministry, who was the first to visit Iraq?"* Reasoning through such questions is very difficult because there are no explicit timestamps provided, requiring extra steps of inference. In Figure 1, LLMs tend to hallucinate when confronted with such questions, leading to incorrect reasoning. Conversely, when temporal events are replaced with specific timestamps, such as *"After 2016-01-05, who was the first to visit Iraq?"*, LLMs can more easily deduce the correct answer. Therefore, we believe that converting implicit questions into explicit ones is a crucial issue.

(2) Lack of Temporal Knowledge. To enhance the reasoning capability of LLMs, previous methods (Li et al., 2023) employ off-the-shelf retrieval tools such as BM25 to extract relevant facts from a knowledge graph as background knowledge. However, these retrieval methods focus solely on semantic matching, thus the retrieved knowledge neglects the time constraints of the question, rendering it ineffective for reasoning. For example, in Figure 1, quadruple *(Evan, visit, Iraq, 2016-01-04)* is irrelevant because the question requires retrieving facts after 2016-01-05. Therefore, constructing a retriever that concurrently pays attention to semantic similarity and temporal constraints is of great importance for the TKGQA task.

To address the above challenges, we propose TimeR^4, a Time-aware Retrieve-Rewrite-Retrieve-Rerank framework for the TKGQA task. Specifically, to mitigate the issue of LLMs hallucinating when faced with implicit temporal questions, we employ the *retrieve-rewrite* strategy. We perform fact retrieval from the Facts Knowledge Store (FKS) to obtain relevant facts for implicit questions. Subsequently, we rewrite these questions by replacing temporal facts with specific timestamps, ensuring that all questions contain explicit temporal information. For constructing FKS, we fix the language model to obtain semantical fact embeddings. To simultaneously capture semantic similarity and temporal constraints, we employ the *retrieve-rerank* strategy. This involves conducting a time-aware retrieval from the Temporal Knowledge Store (TKS) and reranking the facts based on temporal constraints to refine the retrieval process. The TKS is constructed by fine-tuning a language model with a contrastive time-aware retrieval strategy, which develops an encoder capable of capturing both semantic similarity and temporal constraints by constructing three types of negatives for each question. Finally, we fine-tune open-source LLMs with the retrieved facts from TKS, leveraging the temporal knowledge in TKGs to enhance the model's temporal reasoning capabilities. Experiments on two datasets demonstrate that our strategy significantly enhances the temporal reasoning abilities of LLMs. Overall, our work makes the following contributions:

- We integrate LLMs with TKGs and propose TimeR^4, a Time-aware Retrieve-Rewrite-Retrieve-Rerank framework, which effectively overcomes the limitations in handling temporal knowledge in LLMs.
- We propose a contrastive time-aware retrieval strategy that simultaneously pays attention to semantic similarity and temporal constraints.
- The experimental results demonstrate that our approach achieves relative improvements of 47.8% and 22.5% respectively on two TKGQA datasets.

2 Related Work

2.1 KG-enhanced LLMs

Considering the excellent reasoning ability of large language models (LLMs) on NLP tasks (Bang et al., 2023), many recent works have applied LLMs on KGQA tasks. Based on how these methods integrate with knowledge graphs, they can be categorized into three primary types: retrieval-based reasoning, path-based reasoning, and agent-based reasoning. Retrieval-based methods (Baek et al., 2023b; He et al., 2024) focus on retrieving relevant subgraphs or triples from the knowledge graph that contain the information needed to answer a question. The LLMs are then used to process and reason over these retrieved information. However, they only consider semantic similarity and neglect temporal constraints. Path-based methods (Luo et al., 2023; Cheng et al., 2024) involve exploring paths within the knowledge graph to establish connections between the question and the potential answers. These methods typically utilize LLMs to traverse the graph and generate possible paths. However, they cannot be directly applied to TKGQA because they do not account for temporal dimension in path reasoning. Agent-based methods(Sun et al., 2023; Jiang et al., 2023) treat LLMs as an agent to search and prune on the KGs to find answers. However, they prove inefficient for complex reasoning

在 TKGQA 任务中，有许多隐式问题，例如 *"After the Danish Ministry, who was the first to visit Iraq?"*。通过这些问题进行推理非常困难，因为没有提供明确的时间戳，需要额外的推理步骤。在图 1 中，当面对这样的问题时，LLM 往往会产生幻觉，导致推理错误。相反，当将时间事件替换为具体的时间戳，如 *"After 2016-01-05, who was the first to visit Iraq?"* 时，LLM 可以更容易地推导出正确答案。因此，我们认为将隐式问题转换为显式问题是一个关键问题。

(2) 缺乏时间知识。为了增强大型语言模型的推理能力，先前的方法（Li 等, 2023）使用现成的检索工具如 BM25，从知识图谱中提取相关事实作为背景知识。然而，这些检索方法仅关注语义匹配，因此检索到的知识忽略了问题的时间约束，使其在推理中无效。例如，在图 1 中，四元组 *(Evan, visit, Iraq, 2016-01-04)* 是无关的，因为问题要求检索 2016-01-05 之后的事实。因此，构建一个同时关注语义相似性和时间约束的检索器，对于 TKGQA 任务具有重要意义。

为了解决上述挑战，我们提出了TimeR[4]，一种面向时间的检索-重写-再检索-重排序框架，用于TKGQA任务。具体来说，为了减轻LLM在面对隐性时间问题时产生幻觉的问题，我们采用了*retrieve-rewrite*策略。我们从事实知识库（FKS）中检索事实，以获取与隐性问题相关的事实。随后，我们通过用具体时间戳替换时间事实来重写这些问题，确保所有问题都包含明确的时间信息。在构建FKS时，我们固定语言模型以获取语义事实嵌入。为了同时捕捉语义相似性和时间约束，我们采用了*retrieve-rerank*策略。这涉及从时间知识库（TKS）进行时间感知检索，并根据时间约束对事实进行重排序，从而优化检索过程。TKS是通过对语言模型进行对比时间感知微调而构建的。每个问题。最后，我们使用从TKS检索的事实对开源大型语言模型进行微调，利用TKGs中的时间知识来增强模型的时间推理能力。在两个数据集上的实验表明，我们的策略显著提升了大型语言模型的时间推理能力。总体而言，我们的工作做出了以下贡献：

- 我们将大型语言模型（LLM）与时间知识图谱（TKG）整合，并提出了 TimeR[4]，一个时间感知的检索-重写-检索-重排序框架，有效克服了大型语言模型在处理时序知识方面的局限性。
- 我们提出了一种对比时序感知检索策略，同时关注语义相似性和时间约束。
- 实验结果表明，我们的方法在两个TKGQA数据集上分别实现了47.8%和22.5%的相对提升。

2 相关工作

2.1 KG增强的LLM

考虑到大型语言模型（LLMs）在自然语言处理任务上的出色推理能力（Bang 等，2023），许多近期的研究工作已将 LLMs 应用于知识图谱问答（KGQA）任务。根据这些方法与知识图谱的整合方式，它们可以分为三种主要类型：基于检索的推理、基于路径的推理和基于代理的推理。基于检索的方法（Baek 等，2023b；He 等，2024）侧重于从知识图谱中检索相关的子图或三元组，这些子图或三元组包含回答问题所需的信息。然后使用 LLMs 对这些检索到的信息进行处理和推理。然而，它们只考虑语义相似性而忽略了时间约束。基于路径的方法（Luo 等，2023；Cheng 等，2024）涉及在知识图谱中探索路径，以建立问题与潜在答案之间的连接。这些方法通常利用 LLMs 遍历图谱并生成可能的路径。然而，

tasks due to their reliance on multiple LLM-calls. Furthermore, the greedy decision-making process is susceptible to error propagation.

2.2 TKGQA Methods

The TKGQA task is more challenging than the KGQA task due to the added temporal dimension of reasoning. To incorporate temporal information, some methods pose a question as a TKG completion problem and utilize TKG embedding score functions to select entities or timestamps with the highest relevance as answers (Saxena et al., 2021). TempoQR (Mavromatis et al., 2022) augments the question embeddings with context, entity, and time-aware information by three designed modules. MultiQA (Chen et al., 2023b) adopts Transformer encoding layers to aggregate multi-granularity time information. Some works integrate temporal information by introducing RGCN (Relational Graph Convolutional Networks). EXAQT (Jia et al., 2021) utilizes the RGCN layer and augments it with dictionary matching. TwiRGCN (Sharma et al., 2022) adopts temporally weighted graph convolution followed by answer gating. LGQA (Liu et al., 2023) applies a multi-hop message passing graph neural network layer to combine the global and local information. However, such two methods perform poorly in complex reasoning tasks, especially those involving multi-granularity temporal questions. ARI (Chen et al., 2023a) integrates LLMs through a knowledge adaptability framework and abstract methodological guidance. However, it cannot be applied to smaller-scale LLMs.

3 Preliminaries

Temporal knowledge graph $\mathcal{G} = \{\mathcal{E}, \mathcal{P}, \mathcal{T}, \mathcal{F}\}$ is a directed graph where vertices are a set of entities $\mathcal{E}$. The edges are a set of predicates $\in \mathcal{P}$ with timestamps $\mathcal{T}$. The quadruple set $\mathcal{F} = \{(s, p, o, t)\} \subseteq \mathcal{E} \times \mathcal{P} \times \mathcal{E} \times \mathcal{T}$ represents the temporal facts, where s and o are subject and object, p is predicate between s and o at timestamp t.

TKGQA is a task to infer the correct answer to natural language question $q \in Q$ based on relevant quadruples $f = (s, p, o, t)$ in the TKG, where the answer can be either an entity name or timestamp.

4 Method

4.1 Overview

Figure 2 presents our proposed model, TimeR^4, a Retrieve-Rewrite-Retrieve-Rerank framework. The detailed training procedure is illustrated in Algorithm 1. To enhance the performance of LLMs in handling complex problems, we first propose a *retrieve-rewrite* strategy. This strategy aims to retrieve implicit temporal knowledge within the questions from the Facts Knowledge Store (FKS) and reformulate the questions using this background knowledge to include explicit time constraints. For FKS, we utilize the language model to obtain embeddings of facts within TKGs. Next, we implement a *retrieve-rerank* module to retrieve both semantically and temporally relevant facts. This involves conducting a time-aware retrieval from the Temporal Knowledge Store (TKS) and reranking the facts based on temporal constraints. For TKS, we fine-tune a language model using contrastive learning to develop an encoder capable of simultaneously capturing semantic similarity and temporal constraints. Finally, we fine-tune the open-source LLMs, incorporating the retrieved facts from TKGs for enhanced LLMs' temporal reasoning.

Algorithm 1: The training procedure of TimeR^4

Input : TKG $\mathcal{G}$, Questions $\mathcal{Q}$, Ground Trurh gt, Language Model *LM*, raw LLM M

Output : Fine-tuned LLM M'

negatives $\leftarrow$ `GenerateNegatives(`$\mathcal{G}$, $\mathcal{Q}$, gt`)`
$LM_t \leftarrow$ Optimize *LM* as Equation 8
FKS $\leftarrow$ `ConstructKS(`$\mathcal{G}$, *LM*`)`
TKS $\leftarrow$ `ConstructKS(`$\mathcal{G}$, LM_t`)`
for $\{q\} \in loader(\mathcal{Q})$ **do**
  $f \leftarrow$ `Retrieve(`*FKS*, q, *LM*`)`;
  $q^* \leftarrow$ `ReWrite(`q, f`)`;
  $f' \leftarrow$ `Retrieve(`*TKS*, q^*, LM_t`)`;
  $f^+ \leftarrow$ `ReRank(`f', q`)`;
  $M' \leftarrow$ Optimize M as Equation 11
end
Function `Retrieve(`*KS*, q, *LM*`)`**:**
  $\mathbf{E_q} \leftarrow$ `LM(`q`)`;
  $\phi_s \leftarrow$ `cos(`$\mathbf{E_q}$, *KS*`)`;
  $f^+ \leftarrow$ `TopN(`ϕ_s`)`;
  return f^+;
Function `ConstructKS(`$\mathcal{G}$, *LM*`)`**:**
  KS $\leftarrow$ `LM(`$\mathcal{G}$`)`;
  return *KS*;

由于依赖多次大型语言模型调用，这些任务面临挑战。此外，贪婪的决策过程容易导致错误传播。

2.2 TKGQA 方法

由于加入了时间维度的推理，TKGQA 任务比 KGQA 任务更具挑战性。为了融入时间信息，一些方法将问题视为 TKG 完成问题，并利用 TKG 嵌入评分函数选择相关性最高的实体或时间戳作为答案（Saxena 等, 2021）。TempoQR（Mavromatis 等, 2022）通过三个设计的模块，用上下文、实体和时间感知信息增强问题嵌入。MultiQA（Chen 等, 2023b）采用 Transformer 编码层来聚合多粒度时间信息。一些工作通过引入 RGCN（关系图卷积网络）来整合时间信息。EXAQT（Jia 等, 2021）利用 RGCN 层，并通过字典匹配进行增强。TwiRGCN（Sharma 等, 2022）采用时间加权图卷积，然后进行答案门控。LGQA（Liu 等, 2023）应用多跳消息传递图神经网络层进行结合

Algorithm 1: The training procedure of TimeR^4

Input : TKG $\mathcal{G}$, Questions $\mathcal{Q}$, Ground Trurh gt, Language Model *LM*, raw LLM *M*

Output : Fine-tuned LLM M'

negatives $\leftarrow$ `GenerateNegatives(`$\mathcal{G}$, $\mathcal{Q}$, gt`)`

$LM_t \leftarrow$ Optimize *LM* as Equation 8

FKS $\leftarrow$ `ConstructKS(`$\mathcal{G}$, *LM*`)`

TKS $\leftarrow$ `ConstructKS(`$\mathcal{G}$, LM_t`)`

for $\{q\} \in loader(\mathcal{Q})$ **do**

  $f \leftarrow$ `Retrieve(`*FKS*, q, *LM*`)`;

  $q^* \leftarrow$ `ReWrite(`q, f`)`;

  $f' \leftarrow$ `Retrieve(`*TKS*, q^*, LM_t`)`;

  $f^+ \leftarrow$ `ReRank(`f', q`)`;

  $M' \leftarrow$ Optimize *M* as Equation 11

end

Function `Retrieve(`*KS*, q, *LM*`)`**:**

  $\mathbf{E_q} \leftarrow$ `LM(`q`)`;

  $\phi_s \leftarrow$ `cos(`$\mathbf{E_q}$, *KS*`)`;

  $f^+ \leftarrow$ `TopN(`ϕ_s`)`;

  return f^+;

Function `ConstructKS(`$\mathcal{G}$, *LM*`)`**:**

  KS $\leftarrow$ `LM(`$\mathcal{G}$`)`;

  return *KS*;

3 预备事项

时序知识图 $\mathcal{G} = \{\mathcal{E}, \mathcal{P}, \mathcal{T}, \mathcal{F}\}$ 是一个有向图，其中顶点是一组实体 $\mathcal{E}$。边是一组带有时间戳 $\mathcal{T}$的谓词 $\in \mathcal{P}$。四元组集合 $\mathcal{F} = \{(s, p, o, t)\} \subseteq \mathcal{E} \times \mathcal{P} \times \mathcal{E} \times \mathcal{T}$表示时序事实，其中 s 和 o 是主语和宾语，p 是 s 和 o 之间在时间戳 t 的谓词。

TKGQA 是一个任务，用于根据 TKG 中相关的四元组 $f = (s, p, o, t)$ 推断自然语言问题 $q \in Q$ 的正确答案，其中答案可以是实体名称或时间戳。

4 方法

4.1 概述

图2展示了我们提出的模型 TimeR^4，一个检索-重写-再检索-重新排序框架。详细的训练过程在算法1中进行了说明。为了提高大语言模型（LLMs）处理复杂问题的性能，我们首先提出了一种*retrieve-rewrite*策略。该策略旨在从事实知识库（FKS）中检索问题中隐含的时间知识，并利用这些背景知识重构问题，以包含明确的时间约束。对于FKS，我们使用语言模型获取时间知识图（TKGs）中事实的嵌入。接下来，我们实现了一个*retrieve-rerank*模块，以检索语义和时间上相关的事实。这涉及对时间知识库（TKS）进行时间感知检索，并根据时间约束对事实进行重新排序。对于TKS，我们通过对比学习微调语言模型，以开发一个能够同时捕捉语义相似性和时间约束的编码器。最后，我们微调开源大语言模型，结合从TKGs检索到的事实，以增强LLMs的时间推理能力。

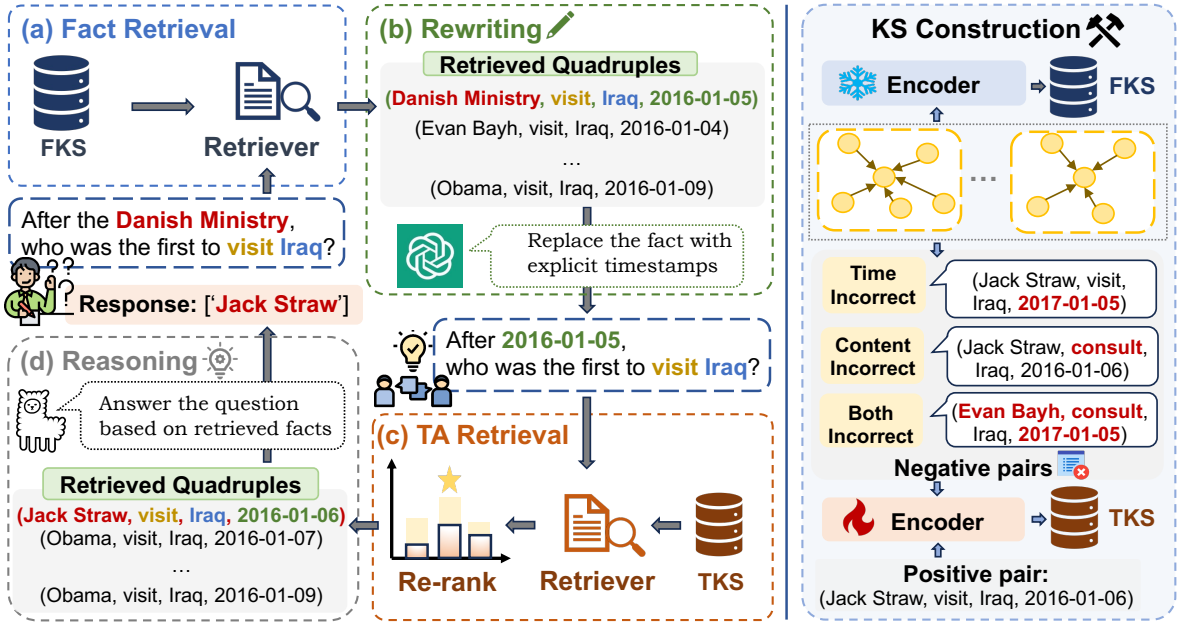


Figure 2: The architecture of TimeR^4 can be divided into four modules, fact retrieval, rewriting, time-aware retrieval, and reasoning. The right part shows how we construct the Knowledge Store (KS).

4.2 Fact Retrieval

Previous works (Chen et al., 2023b; Liu et al., 2023) employed entity-linking tools to identify entities and relations in question, subsequently utilizing these entities and relations for further retrieval. However, some TKGs, such as ICEWS (García-Durán et al., 2018a), lack entity-linking tools, resulting in poor performance with such methods (Chen et al., 2023b). Motivated by the recent study (Baek et al., 2023a), we adopt a direct fact retrieval strategy from TKGs without entity linking.

We first convert each quadruple into natural language sequences. Then we embed all quadruples $T(s, p, o, t) \in \mathcal{G}$ in TKGs onto a dense embedding space by using a pre-trained language model as in Equation 1 and memorize the knowledge representations in a Fact Knowledge Store (FKS). We also embed the given questions q as in Equation 2. d is the dimension of the output vector.

$$\text{FKS} = \{\mathbf{E}_f | \mathbf{E}_f = LM(S(s, p, o, t)), (s, p, o, t) \in \mathcal{G}\} \quad (1)$$

$$\mathbf{E}_q = LM(q) \in \mathbb{R}^d \quad (2)$$

To retrieve the k-nearest semantic quadruples $\mathbf{f}_1^+$ according to the representation distance for the given question, we calculate the similarity between $\mathbf{E}_q$ and $\mathbf{E}_f$. We use the FAISS library (Johnson et al., 2021) for indexing and similarity calculation.

$$\phi_{FKS}(\mathbf{E}_q, \mathbf{E}_t) = \cos(\mathbf{E}_q, \mathbf{E}_f) = \mathbf{E}_q \cdot \mathbf{E}_f \quad (3)$$

$$\mathbf{f} = \arg\max \phi_{FKS}(\mathbf{E}_q, \mathbf{E}_f) \quad (4)$$

4.3 Rewrite

Complex questions often contain implicit temporal information, posing a challenge to the TKGQA task. To address the hallucination issues of LLMs with implicit questions, we plan to rewrite the questions to ensure that all questions have explicit timestamps. Specifically, we retrieve the necessary background facts through the FKS and then input them along with the question into the LLM for inference and rewriting as in Equation 5. We apply the in-context learning strategy (Dong et al., 2023), which encodes structural knowledge into demonstrations to guide the LLM. The specific prompt is shown in Appendix A.

$$q^* = LLM(Prompt(q, f)), q \in Q \quad (5)$$

After being rewritten by the LLMs, questions containing implicit temporal facts are modified to include explicit timestamps based on the retrieved background facts. Additionally, for certain questions involving common knowledge not present in TKGs, such as *"Who was the first president of the US after World War II?"*, the LLMs can utilize their inherent knowledge to rewrite them. Knowing that World War II ended in 1945, the question can successfully be transformed into *"Who was the first president of the US after 1945?"*.

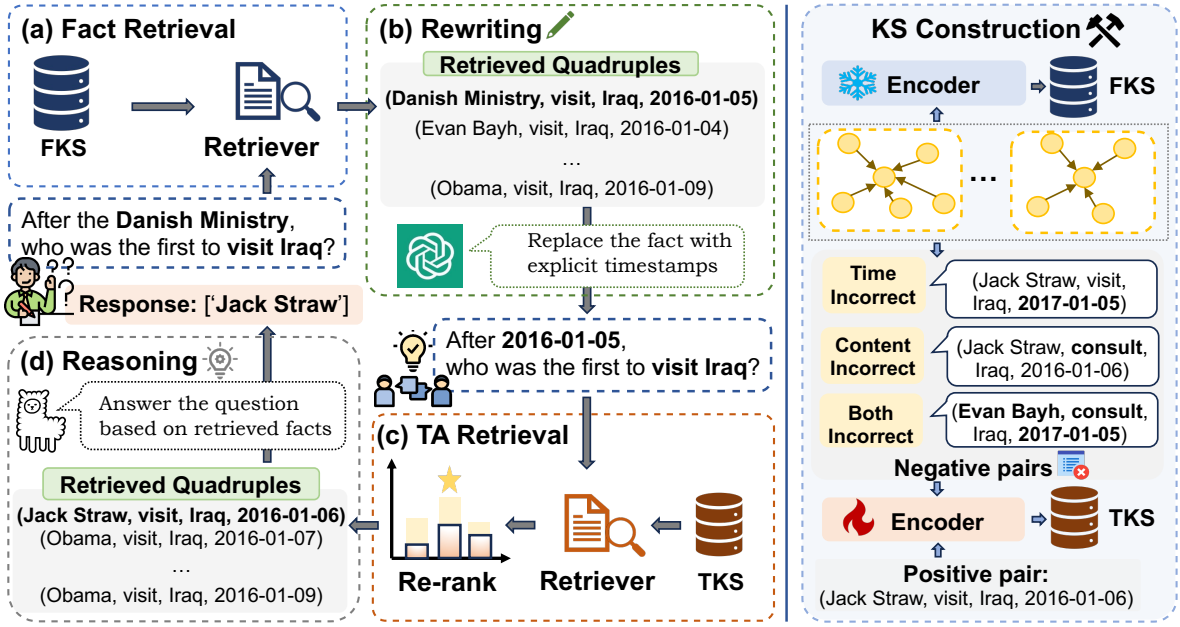


图 2：TimeR^4 的架构可以分为四个模块：事实检索、重写、时间感知检索和推理。右边部分展示了我们如何构建知识存储（KS）。

4.2 事实检索

以往的工作（Chen 等, 2023b；Liu 等, 2023）使用实体链接工具来识别问题中的实体和关系，随后利用这些实体和关系进行进一步检索。然而，一些时态知识图（TKG），如 ICEWS（García-Durán 等, 2018a），缺乏实体链接工具，导致此类方法的表现不佳（Chen 等, 2023 b）。受到近期研究（Baek 等, 2023a）的启发，我们采用从 TKG 中直接检索事实的策略，而无需实体链接。

我们首先将每个四元组转换为自然语言序列。然后我们将时序知识图谱（TKGs）中的所有四元组 $T(s,p,o,t) \in \mathcal{G}$ 嵌入到稠密嵌入空间中，使用如公式 1 所示的预训练语言模型，并将知识表示存储在事实知识库（FKS）中。我们还按照公式 2 将给定的问题 q 进行嵌入。d 是输出向量的维度。

$$\text{FKS} = \{\mathbf{E}_f | \mathbf{E}_f = LM(S(s,p,o,t)), (s,p,o,t) \in \mathcal{G}\} \quad (1)$$

$$\mathbf{E}_q = LM(q) \in \mathbb{R}^d \quad (2)$$

要根据给定问题按照表示距离检索 k 个最近的语义四元组 $\mathbf{f}_1^+$，我们计算 $\mathbf{E}_q$ 与 $\mathbf{E}_f$ 之间的相似性。我们使用 FAISS 库（Johnson 等, 2021）进行索引和相似性计算。

$$\phi_{FKS}(\mathbf{E}_q, \mathbf{E}_t) = \cos(\mathbf{E}_q, \mathbf{E}_f) = \mathbf{E}_q \cdot \mathbf{E}_f \quad (3)$$

$$\mathbf{f} = \arg\max \phi_{FKS}(\mathbf{E}_q, \mathbf{E}_f) \quad (4)$$

4.3 重写

复杂问题通常包含隐含的时间信息，这对 TKGQA 任务构成了挑战。为了解决 LLM 在处理隐含问题时的幻觉问题，我们计划重写问题，以确保所有问题都有明确的时间戳。具体来说，我们通过 FKS 检索必要的背景事实，然后将这些事实与问题一起输入 LLM 进行推理和重写，如公式 5 所示。我们应用了上下文学习策略（Dong 等, 2023），该策略将结构化知识编码到示例中以指导 LLM。具体提示见附录 A。

$$q^* = LLM(Prompt(q, f)), q \in Q \quad (5)$$

在被大语言模型重写后，包含隐含时间事实的问题会被修改为根据检索到的背景事实加入明确的时间戳。此外，对于某些涉及 TKGs 中不存在的常识性问题，例如 *"Who was the first president of the US after World War II?"*，大语言模型可以利用其固有知识进行重写。了解到第二次世界大战于1945年结束，该问题可以成功地被转化为 *"Who was the first president of the US after 1945?"*。

4.4 Time-aware Retrieval

We further propose the Time-aware Retrieval module to retrieve facts that satisfy both semantic similarity and time constraints simultaneously from the temporal knowledge store (TKS). To construct the TKS, we fine-tuned the language model to model the time-aware facts and store them in the knowledge base as in Equation 6.

$$\text{TKS} = \{\mathbf{E}_t | \mathbf{E}_t = LM_t(S(s,p,o,t)), (s,p,o,t) \in \mathcal{G}\} \quad (6)$$

In order to enhance the time-awareness of the language model, we employ the contrastive time-aware retrieval strategy. We randomly corrupt the time, relations, and entities of the positive pair separately and generate three types of negative pairs: time incorrect, content incorrect, and both incorrect, as shown in Figure 2. Contrastive loss is calculated based on the cosine similarity between the question representation $\mathbf{E}_q$ and the quadruples representation $\mathbf{E}_t \in$ TKS as Equation 7.

$$\phi_{TKS} = \cos(\mathbf{E}_{q^*}, \mathbf{E}_t) \quad (7)$$

We aim to minimize the distance between positive pairs and maximize the distance between negative pairs. The contrastive label Y=1 signifies that the context corresponds to a positive pair of the question, whereas Y=0 indicates that the context represents the negative pair. Subsequently, following Son and Oh (2023), the contrastive loss is computed in Equation 8, with w_p and w_n as the weights for positive and negative samples.

$$\mathcal{L} = \sum_i [w_p Y \cdot \exp(\phi_{TKS}) + w_n (1-Y) \cdot \exp(1-\phi_{TKS})] \quad (8)$$

4.5 Rerank

To further refine the retrieval process, we design a time-filtering function to filter out irrelevant facts and focus more on time-related ones. After the rewriting module, each question q^* contains a specific timestamp t_q, except for the questions that require answering the timestamp. In that case, to introduce the influence of time intervals, we design a time-filtering function for questions containing time constraints. For each quadruple (s,p,o,t) in TKG $\mathcal{G}$, we calculate the time difference between t_q and t, filter out quadruples that fall outside the range, and normalize the time differences within the range as the results of time filtering function. Equation 9 represents the time-filtering function for "before" type questions.

$$\phi_t(t_q, t) = \begin{cases} 1 - \frac{|t_q - t|}{max(t_q - t)}, & \text{if } (t_q - t) > 0 \\ -100, & \text{otherwise} \end{cases} \quad (9)$$

Specifically, we add the score obtained from the time filtering function $\phi_t(t_q, t)$ to the score of the Time-aware Retirver $\phi_{TKS}(\mathbf{E}_q, \mathbf{E}_t)$, thereby reranking the scores of the retrieved facts. μ is the weight of two scores.

$$\phi(q,t) = \mu \cdot \phi_{TKS}(\mathbf{E}_{q^*}, \mathbf{E}_t) + (1-\mu) \cdot \phi_t(t_q, t) \quad (10)$$

4.6 Reasoning

After obtaining the retrieved quadruples we formulate the reasoning part as an LLM optimization problem, aiming to maximize the probability of inferring the answer a to question q from the knowledge graph $\mathcal{G}$ by using the retrieved quadruples f^+ as history facts as in Equation 11. To guide the LLM in generating final answers, we design a simple instruction prompt in Appendix A.

$$\mathcal{L} = \max_{\Phi} \sum_{(q^*,a) \in \hat{\mathcal{Q}}} \sum_{t=1}^{|a|} \log \left(P_\Phi \left(a_t \mid (q^*, f^+), a_{<t} \right) \right) \quad (11)$$

5 Experiments

5.1 Experimental Settings

Datasets. Considering that the CronQuestions (Saxena et al., 2021) dataset has been reported to contain spurious correlations that different models can exploit to achieve high accuracy (Sharma et al., 2022), we base our experiments on two recent more challenging datasets, i.e., MULTITQ (Chen et al., 2023b) and TimeQuestions (Jia et al., 2021). **MULTITQ** is the largest known TKGQA dataset constructed from the ICEWS05-15 (García-Durán et al., 2018b), which has 500K unique question-answer pairs. Besides, MULTITQ features multiple temporal granularities, including years, months, and days, with questions spanning over 3600 days. **TimeQuestions** is another challenging dataset, which has 16K temporal questions and is divided into four categories. However, TimeQuestions only includes a time granularity of years and is much smaller in size. The statistical information is presented in Table 1 and Table 2 separately.

Baseline. We select three types of baselines for comparison on MULTITQ: (1) Pre-trained LMs, including BERT (Devlin et al., 2019) and ALBERT (Lan et al., 2020). (2) Embedding-based methods, including EmbedKGQA (Saxena et al.,

4.4 时间感知检索

我们进一步提出了时间感知检索模块，以从时间知识存储（TKS）中同时检索满足语义相似性和时间约束的事实。为了构建TKS，我们对语言模型进行了微调，以对时间感知事实进行建模，并将它们存储在知识库中，如公式6所示。

$$\text{TKS} = \{\mathbf{E}_t | \mathbf{E}_t = LM_t(S(s,p,o,t)), (s,p,o,t) \in \mathcal{G}\} \quad (6)$$

为了增强语言模型的时间感知能力，我们采用对比时间感知检索策略。我们分别随机篡改正样本对的时间、关系和实体，并生成三种类型的负样本对：时间错误、内容错误以及两者均错误，如图 2 所示。对比损失是基于问题表示 $\mathbf{E}_q$ 与四元组表示 $\mathbf{E}_t \in$ 的余弦相似度计算的，如公式 7 所示。

$$\phi_{TKS} = \cos(\mathbf{E}_{q^*}, \mathbf{E}_t) \quad (7)$$

我们的目标是最小化正样本对之间的距离，并最大化负样本对之间的距离。对比标签 Y=1 表示上下文对应于问题的正样本对，而 Y=0 表示上下文代表负样本对。随后，按照 Son 和 Oh (2023) 的方法，对比损失在公式 8 中计算，其中 w_p 和 w_n 分别为正样本和负样本的权重。

$$\mathcal{L} = \sum_i [w_p Y \cdot \exp(\phi_{TKS}) + w_n (1-Y) \cdot \exp(1-\phi_{TKS})] \quad (8)$$

4.5 重新排序

为了进一步优化检索过程，我们设计了一个时间过滤功能，以滤除无关的事实，更关注与时间相关的事实。在重写模块之后，每个问题 q^* 都包含一个特定的时间戳 t_q，除了那些需要回答时间戳的问题。在这种情况下，为了引入时间间隔的影响，我们为包含时间约束的问题设计了时间过滤功能。对于 TKG $\mathcal{G}$ 中的每个四元组 (s,p,o,t)，我们计算 t_q 与 t 之间的时间差，过滤掉超出范围的四元组，并将范围内的时间差归一化作为时间过滤功能的结果。公式 9 表示“之前”类型问题的时间过滤功能。

$$\phi_t(t_q, t) = \begin{cases} 1 - \frac{|t_q - t|}{max(t_q - t)}, & \text{if } (t_q - t) > 0 \\ -100, & \text{otherwise} \end{cases} \quad (9)$$

具体来说，我们将时间过滤函数 $\phi_t(t_q, t)$ 得到的分数加到时间感知检索器 $\phi_{TKS}(\mathbf{E}_q, \mathbf{E}_t)$ 的分数上，从而重新排序检索到的事实的分数。μ 是两个分数的权重。

$$\phi(q,t) = \mu \cdot \phi_{TKS}(\mathbf{E}_{q^*}, \mathbf{E}_t) + (1-\mu) \cdot \phi_t(t_q, t) \quad (10)$$

4.6 推理

在获取检索到的四元组后，我们将推理部分表述为一个大语言模型（LLM）优化问题，旨在通过使用检索到的四元组 f^+ 作为历史事实，从知识图谱 $\mathcal{G}$ 中推断问题 q 的答案 a 的概率最大化，如公式 11 所示。为了引导 LLM 生成最终答案，我们在附录 A 中设计了一个简单的指令提示。

$$\mathcal{L} = \max_{\Phi} \sum_{(q^*,a) \in \hat{\mathcal{Q}}} \sum_{t=1}^{|a|} \log \left(P_\Phi \left(a_t \mid (q^*, f^+), a_{<t} \right) \right) \quad (11)$$

5 个实验

5.1 实验设置

数据集。考虑到CronQuestions（Saxena 等，2021）数据集据报道包含虚假相关性，不同模型可以利用这些相关性来实现高准确率（Sharma 等，2022），我们在两个最近更具挑战性的数据集上进行实验，即MULTITQ（Chen 等，2023b）和TimeQuestions（Jia 等，2021）。MULTITQ是已知最大的基于ICEWS05-15（García-Durán 等，2018b）构建的时序知识图问答（TKGQA）数据集，包含50万个唯一问答对。此外，MULTITQ具有多种时间粒度，包括年、月和日，问题跨越超过3600天。TimeQuestions是另一个具有挑战性的数据集，包含1.6万个时序问题，并分为四类。然而，TimeQuestions仅包含年度时间粒度，规模也小得多。统计信息分别呈现在表1和表2中。

基线。我们选择三种类型的基线在 MULTITQ 上进行比较：（1）预训练语言模型，包括 BERT（Devlin 等人，2019）和 ALBERT（Lan 等人，2020）。（2）基于嵌入的方法，包括 EmbedKGQA（Saxena 等人，

Category		Train	Dev	Test
Single	Equal	135,890	18,983	17,311
	Before/After	75,340	11,655	11,073
	First/Last	72,252	11,097	10,480
Multiple	Equal Multi	16,893	3,213	3,207
	After First	43,305	6,499	6,266
	Before Last	43,107	6,532	6,247
Total		386,787	587,979	54,584

Table 1: Statistics of MULTITQ dataset.

Category	Train	Dev	Test
Explicit	2,724	1,302	1,311
Implicit	651	291	292
Temporal	2,657	1,073	1,067
Ordinal	938	570	567
Total	6,970	3,236	3,237

Table 2: Statistics of TIMEQUESTIONS dataset.

2020), CronKGQA (Saxena et al., 2021), MultiQA (Chen et al., 2023b). (3) LLM-based methods, including ARI (Chen et al., 2023a), LLaMA2 (Touvron et al., 2023), ChatGPT. For TimeQuestions, we also use three types of baselines: (1) KGQA method, including PullNet (Sun et al., 2019), Uniqorn (Pramanik et al., 2023), and GRAFT-Net (Sun et al., 2018). (2) TKGQA methods, including CronKGQA, TempoQR (Mavromatis et al., 2022), EXAQT (Jia et al., 2021), LGQA (Liu et al., 2023), and TwiRGCN (Sharma et al., 2022). (3) LLM-based methods, including LLaMA2 and ChatGPT. We only input the given questions into ChatGPT and LLaMA2 without any explanation.

Implementations Details. We use LLaMA2-Chat-7B (Touvron et al., 2023) as the LLM backbone. We fine-tune the LLaMA2 for 2 epochs on 2 NVIDIA A6000 GPUs. We only use 20% of the training data for MULTITQ datasets for training because it is very large. For Fact Retriever, We utilize off-the-shell SentenceBert (Reimers and Gurevych, 2019) as the base encoder. For Time-aware Retriever, we fine-tune the SentenceBert for 10 epochs. For re-writing, we use the OpenAI-API [1] (gpt-3.5-turbo-0125[2]). We set the μ as 0.4. We fine-tune each question using in-batch negatives and three hard negatives.

5.2 Main Results

We present the experimental results in comparison with other methods of TimeR^4 on the MultiTQ and TimeQuestions datasets in Table 3 and Table 4, where the highest results are highlighted in bold font and the second highest results are marked underlined. TimeR^4 achieves the best performance in all experimental settings, indicating its superiority on the TKGQA task.

For the MultiTQ dataset, TimeR^4 achieves state-of-the-art performance across all question types. Specifically, We find that PLMs (BERT. ALBERT) and LLMs (LLaMA2, ChatGPT) exhibit the lowest performance on the TKGQA task. This might be due to the lack of necessary temporal knowledge, thus leading to errors in reasoning. Compared to traditional methods, TimeR^4 shows a significant improvement in hits@1 performance, achieving a 59.8% enhancement. This highlights the capability of LLMs in reasoning on complex temporal questions, particularly those involving multi-granularity timestamps and complex reasoning. Furthermore, compared with the recent ChatGPT-based method ARI, TimeR^4 demonstrates a 47.8% improvement, showcasing the effectiveness of our proposed temporal reasoning framework.

As for TimeQuestions, TimeR^4 also achieves the best results across all question types. The results of KGQA methods are the poorest for they lack the ability to retrieve temporal facts or reason in temporal facts. Compared to traditional methods, TimeR^4 still achieves a 22.5% relative improvement on Hits@1. The results demonstrate the capability of TimeR^4 for precisely answering temporal questions. Compared with other LLMs, TimeR^4 also performs the best. The results of ChatGPT and LLaMA2 are much higher than on MULTITQ datasets. This might be because Timequestions are built on the Wikidata (Vrandečić and Krötzsch, 2014) knowledge graph, and most LLMs are pre-trained on the Wikidata knowledge graph corpus. Therefore, they store some relevant information, enabling them to answer such questions.

5.3 Ablation Study

The ablation results are shown in Table 5. The results for TimeQuestions do not decline significantly, primarily because the MULTITQ dataset is more challenging. MULTITQ is larger and covers a wider time range, making it harder to retrieve relevant facts. Conversely, in the TimeQuestions dataset, relevant facts can often be retrieved by the fact retrieval module, which is why our strategy is less prominent on this dataset.

Effect of fact retrieval module. We first replace fact retriever with entity-linking tools. Since no

[1] https://platform.openai.com/docs/api-reference
[2] https://platform.openai.com/docs/models/gpt-3-5-turbo

Category		Train	Dev	Test
Single	Equal	135,890	18,983	17,311
	Before/After	75,340	11,655	11,073
	First/Last	72,252	11,097	10,480
Multiple	Equal Multi	16,893	3,213	3,207
	After First	43,305	6,499	6,266
	Before Last	43,107	6,532	6,247
Total		386,787	587,979	54,584

表 1：MULTITQ 数据集统计。

Category	**Train**	**Dev**	**Test**
Explicit	2,724	1,302	1,311
Implicit	651	291	292
Temporal	2,657	1,073	1,067
Ordinal	938	570	567
Total	6,970	3,236	3,237

表 2：TIMEQUESTIONS 数据集的统计信息。

2020), CronKGQA (Saxena 等, 2021), MultiQA (Chen 等, 2023b). (3) 基于大语言模型的方法, 包括 ARI (Chen 等, 2023a), LLaMA2 (Touvron 等, 2023), ChatGPT。对于时间类问题，我们还使用三类基线方法: (1) KGQA 方法，包括 Pull Net (Sun 等, 2019), Uniqorn (Pramanik 等, 2023), 以及 GRAFT-Net (Sun 等, 2018)。(2) TKGQA 方法，包括 CronKGQA, TempoQR (Mavromatis 等, 2022), EXAQT (Jia 等, 2021), LGQA (Liu 等, 2023), 以及 TwiRGCN (Sharma 等, 2022)。(3) 基于大语言模型的方法，包括 LLaMA2 和 ChatGPT。我们仅将给定的问题输入到 ChatGPT 和 LLaMA2 中，不提供任何解释。

实现细节。我们使用 LLaMA2-Chat-7B (Touvron 等, 2023) 作为 LLM 骨干。我们在 2 张 NVIDIA A6000 GPU 上对 LLaMA2 进行了 2 个周期的微调。由于 MULTITQ 数据集非常大，我们仅使用 20% 的训练数据进行训练。对于事实检索器，我们使用现成的 SentenceBert (Reimers 和 Gurevych, 2019) 作为基础编码器。对于时间感知检索器，我们对 SentenceBert 进行了 10 个周期的微调。对于重写，我们使用 OpenAI-API [1] (gpt-3.5-turbo-0125[2])。我们将 μ 设置为 0.4。我们使用批内负样本和三个难负样本对每个问题进行了微调。

5.2 主要结果

我们在表 3 和表中展示了 TimeR^4 在 MultiTQ 和 TimeQuestions 数据集上与其他方法的实验结果比较

[1]https://platform.openai.com/docs/api-reference

[2]https://platform.openai.com/docs/models/gpt-3-5-turbo

4，其中最高的结果以加粗字体突出显示，第二高的结果用下划线标记。TimeR^4 在所有实验设置中都取得了最佳性能，表明其在 TKGQA 任务上的优越性。

对于 MultiTQ 数据集，TimeR^4 在所有问题类型上都达到了最先进的性能。具体来说，我们发现 PLMs（BERT、ALBERT）和 LLMs（LLaMA2、ChatGPT）在 TKGQA 任务上的表现最差。这可能是由于缺乏必要的时间知识，从而导致推理错误。与传统方法相比，TimeR^4 在 hits@1 性能上表现出显著提升，实现了 59.8% 的提升。这突显了 LLM 在复杂时间性问题推理方面的能力，特别是涉及多粒度时间戳和复杂推理的问题。此外，与近期基于 ChatGPT 的方法 ARI 相比，TimeR^4 展示了 47.8% 的提升，展示了我们提出的时间推理框架的有效性。

关于时间问题，TimeR^4 在所有问题类型中也取得了最佳结果。KGQA 方法的结果最差，因为它们缺乏检索时间事实或进行时间事实推理的能力。与传统方法相比，TimeR^4 在 Hits@1 上仍然实现了 22.5% 的相对提升。结果展示了 TimeR^4 精确回答时间问题的能力。与其他大语言模型相比，TimeR^4 也表现最佳。ChatGPT 和 LLaMA2 的结果远高于 MULTITQ 数据集上的表现。这可能是因为 Timequestions 构建于 Wikidata (Vrandečić 和 Krötzsch, 2014) 知识图谱上，而大多数大语言模型在 Wikidata 知识图谱语料上进行了预训练。因此，它们存储了一些相关信息，使它们能够回答此类问题。

5.3 消融研究

消融实验结果如表5所示。TimeQuestions 的结果没有显著下降，主要是因为 MULTITQ 数据集更具挑战性。MULTITQ 更大，覆盖的时间范围更广，使得检索相关事实更加困难。相反，在 TimeQuestions 数据集中，相关事实往往可以通过事实检索模块检索到，这也是我们的策略在该数据集上不那么明显的原因。

事实检索模块的效果。我们首先用实体链接工具替换事实检索器。由于没有

Model	Overall	Question Type		Answer Type	
		Single	Multiple	Entity	Time
BERT	8.3	9.2	6.1	10.1	4
ALBERT	10.8	11.6	8.6	13.9	3.2
EmbedKGQA	20.6	23.5	13.4	29	0.1
CronKGQA	27.9	13.4	13.4	32.8	15.6
MultiQA	29.3	34.7	15.9	34.9	15.7
ARI	38.0	68.0	21.0	39.0	34.0
LLaMA2	18.5	22.0	10.1	23.9	5.5
ChatGPT	10.2	14.7	7.7	13.7	2
TimeR^4	**72.8**	**88.7**	**33.5**	**63.9**	**94.5**

Table 3: Performance comparison of different models (in percentage) on MUULTITQ.

Model	Overall	Question Type			Answer Type	
		Explicit	Implicit	Temporal	Ordinal	
PullNet	10.5	2.2	8.1	23.4	2.9	
Uniqorn	33.1	31.8	31.6	39.2	20.2	
GRAFT-Net	45.2	44.5	42.8	51.5	32.2	
CronKGQA	46.2	46.6	44.5	51.1	36.9	
TempoQR	41.6	46.5	3.6	40	34.9	
EXAQT	57.2	56.8	51.2	64.2	42	
TwiRGCN	60.5	60.2	58.6	64.1	51.8	
LGQA	52.9	53.2	50.6	60.5	40.2	
LLaMA2	27.1	26.8	32.5	27.9	23.4	
ChatGPT	45.9	43.3	51.1	46.5	48.1	
GenTKGQA	58.4	59.6	61.1	56.3	57.8	
TimeR^4	**78.1**	**82.3**	**73.0**	**83.0**	**64.9**	

Table 4: Performance comparison of different models (in percentage) on TimeQuestions.

linking tool is available for the ICEWS, we adopted the approach outlined in Chen et al. (2023b), employing a NER tool to identify entities. It can be observed that after replacing the fact retriever module, the overall performances decrease by 13.2% on hits@1, which indicates that the fact retriever module achieves accurate fact recognition.

Effect of rewriting strategy. To verify the role of the rewriting strategy, we then conduct an experiment where we removed the rewriting strategy and solely relied on the original questions and retrieved facts. The results showed a significant decrease, indicating that the rewriting strategy effectively aids LLMs in mitigating the hallucination of implicit temporal questions.

Effect of time-aware retrieval module. We conducted an experiment where we replaced the time-aware retriever with the fact retriever. The results show a significant decline, indicating that our proposed time-aware retriever method is inherently time-aware and performs better than the non-time-aware fact retriever.

Model	Hit@1	
	MULTITQ	TimeQuestions
TimeR^4	**72.78**	**78.1**
w/o fact retrieval	41.04	54.3
w/o rewrite	61.12	77.2
w/o temporal retrieval	70.34	77.3
w/o rerank	63.04	77.9

Table 5: Results of the ablation study. "w/o" means removing the module.

Effect of reranking strategy. Removing the rerank strategy resulted in a significant decrease in model performance, indicating that filtering out irrelevant time information is indeed crucial.

5.4 Further Discussion

To further analyze the superior performance of TimeR^4 compared to other models, we conduct a series of experiments to gain further insights.

5.4.1 Comparison with LLMs

We compare TimeR^4 with other LLMs in both datasets in Table 6. LLMs *w/ TimeR*^{4} indicates that we input the facts retrieved by our strategy and rewritten questions into the LLMs. LLMs *w/ fine-tuned* indicates that we fine-tuned the LLMs with only questions.

First, the results on two LLMs show that with the enhancement of the facts retrieved from TKGs and our retrieve-rewrite-retrieve-rerank strategy, LLMs *w/ TimeR*^{4} have significantly better performance. This suggests that the LLMs possess some degree of simple temporal reasoning capability. However, for more precise and effective reasoning in complex temporal questions, TimeR^4 effectively overcomes the limitations in handling and interpreting time-sensitive knowledge.

To further explore the effectiveness of our method, we also compared the results of LLama with fine-tuned and without fine-tuned. It can be found that the results of LLaMA2 with fine-tuning almost doubled, indicating that fine-tuning can effectively regulate the output of LLMs and help LLMs learn patterns of temporal reasoning, thereby enhancing the results.

It is worth noting that in multiple questions, ChatGPT performs better than TimeR^4. This is mainly because ChatGPT tends to provide all possible answers. For example, in Table 7, for questions with only one entity as the answer, ChatGPT often returns a list of relevant results, which can result in higher performance.

Model	Overall	Question Type		Answer Type	
		Single	Multiple	Entity	Time
BERT	8.3	9.2	6.1	10.1	4
ALBERT	10.8	11.6	8.6	13.9	3.2
EmbedKGQA	20.6	23.5	13.4	29	0.1
CronKGQA	27.9	13.4	13.4	32.8	15.6
MultiQA	29.3	34.7	15.9	34.9	15.7
ARI	38.0	68.0	21.0	39.0	34.0
LLaMA2	18.5	22.0	10.1	23.9	5.5
ChatGPT	10.2	14.7	7.7	13.7	2
TimeR^4	**72.8**	**88.7**	**33.5**	**63.9**	**94.5**

表 3：不同模型在 MUULTITQ 上的性能比较（百分比）

Model	Overall	Question Type			Answer Type
		Explicit	Implicit	Temporal	Ordinal
PullNet	10.5	2.2	8.1	23.4	2.9
Uniqorn	33.1	31.8	31.6	39.2	20.2
GRAFT-Net	45.2	44.5	42.8	51.5	32.2
CronKGQA	46.2	46.6	44.5	51.1	36.9
TempoQR	41.6	46.5	3.6	40	34.9
EXAQT	57.2	56.8	51.2	64.2	42
TwiRGCN	60.5	60.2	58.6	64.1	51.8
LGQA	52.9	53.2	50.6	60.5	40.2
LLaMA2	27.1	26.8	32.5	27.9	23.4
ChatGPT	45.9	43.3	51.1	46.5	48.1
GenTKGQA	58.4	59.6	61.1	56.3	57.8
TimeR^4	**78.1**	**82.3**	**73.0**	**83.0**	**64.9**

表 4：不同模型在 TimeQuestions 上的性能比较（以百分比表示）。

ICEWS 的链接工具可用，我们采用了 Chen 等人 (2023b) 中概述的方法，使用 NER 工具来识别实体。可以观察到，在替换事实检索模块之后，整体表现下降了 13.2% 的 hits@1，这表明事实检索模块实现了准确的事实识别。

重写策略的效果。为了验证重写策略的作用，我们随后进行了一项实验，在实验中我们移除了重写策略，仅依赖原始问题和检索到的事实。结果显示显著下降，这表明重写策略能够有效帮助大语言模型减轻隐含时间问题的虚构现象。

时间感知检索模块的影响。我们进行了一项实验，将时间感知检索器替换为事实检索器。结果显示性能显著下降，表明我们提出的时间感知检索方法本质上具有时间感知能力，并且表现优于非时间感知的事实检索器。

Model	Hit@1	
	MULTITQ	TimeQuestions
TimeR^4	**72.78**	**78.1**
w/o fact retrieval	41.04	54.3
w/o rewrite	61.12	77.2
w/o temporal retrieval	70.34	77.3
w/o rerank	63.04	77.9

表5：消融实验结果。“w/o”表示移除该模块。

重排序策略的效果。移除重排序策略导致模型性能显著下降，这表明过滤掉无关的时间信息确实至关重要。

5.4 进一步讨论

为了进一步分析 TimeR^4 相对于其他模型的优越表现，我们进行了系列实验以获得更多见解。

5.4.1 与大型语言模型的比较

我们在表6中将 TimeR^4 与其他 LLMs 在两个数据集上进行了比较。LLMs *w/ TimeR*^{4} 表示我们将通过我们的策略检索到的事实和重写的问题输入到 LLMs 中。LLMs *w/ fine- tuned* 表示我们仅用问题对 LLMs 进行了微调。

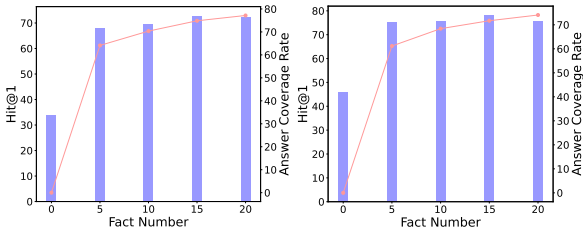
首先，在两个大型语言模型上的结果显示，随着从TKGs获取的事实增强以及我们检索-重写-再次检索-重新排序策略的应用，LLMs *w/ TimeR*^{4} 的性能显著提升。这表明这些LLM具备一定程度的简单时间推理能力。然而，对于更精确和有效的复杂时间问题推理，TimeR^{4}能有效克服在处理和解释时间敏感知识方面的局限性。

为了进一步探讨我们方法的有效性，我们还比较了经过微调和未经过微调的LLama的结果。可以发现，经过微调的LLaMA2的结果几乎翻了一番，这表明微调可以有效地调节大型语言模型的输出，并帮助大型语言模型学习时间推理的模式，从而提升结果。

值得注意的是，在多个问题中，ChatGPT 的表现优于 TimeR^4。这主要是因为 ChatGPT 倾向于提供所有可能的答案。例如，在表 7 中，对于只有一个实体作为答案的问题，ChatGPT 常常返回相关结果的列表，这可能导致更高的表现。

Model	MULTITQ					Timequestions				
	Overall	Question Type		Answer Type		Overall	Question Type		Answer Type	
		Single	Multiple	Entity	Time		Explicit	Implicit	Temporal	Ordinal
TimeR^4	**72.8**	**88.7**	33.5	**63.9**	**94.5**	**78.1**	**82.3**	**73.0**	**83.0**	**64.9**
LLaMA2	18.5	22.0	10.1	23.9	5.5	28.9	26.8	41.9	33.7	33.8
LLaMA2 *w/ finetuned*	33.9	38.4	22.7	45.0	7.8	45.8	44.4	46.0	51.9	37.8
LLaMA2 *w/ TimeR*^{4}	39.1	44.2	26.6	37.0	44.2	59.3	57.4	51.5	73.4	41.0
ChatGPT	10.2	14.7	7.7	13.7	2	45.9	43.3	51.1	46.5	42.1
ChatGPT *w/ TimeR*^{4}	41.4	58.5	**41.2**	56.1	57.1	64.8	66.0	52.9	77.6	45.5

Table 6: Effects of integrating the TimeR^4 framework with different LLMs for reasoning.



(a) Results on MULTITQ. (b) Results on TimeQuestions.

Figure 3: The Hits@1 results of different fact numbers.

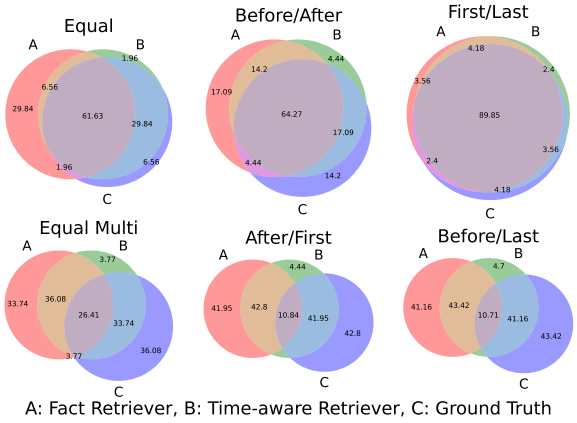


Figure 4: Venn diagrams for the answers coverage overlap of time-aware retrieval, fact retrieval, and ground truth for six question types in MULTITQ dataset.

5.4.2 Number of Retrieved Facts

To explore the impact of varying numbers of retrieved facts on the results, we present the performance changes and the corresponding answer coverage on two datasets by adjusting the number of retrieved facts n in Figure 3. It is evident that the model achieves its peak performance with 15 relevant facts, the same number included in our strategy. Interestingly, there is a slight performance dip at $n = 20$, despite the larger answer coverage obtained with 20 retrieved facts. This suggests that an excessive amount of facts may hinder performance by introducing noise, thereby making it more difficult for the LLMs to distinguish relevant information from irrelevant ones. On the other hand, fewer facts do not provide sufficient temporal knowledge, also leading to poor performance. Hence, $n = 15$ represents the optimal choice.

5.4.3 Effectiveness of the Retrieval

To verify the effectiveness of our time-aware retrieval module, we investigate the overlap of answer coverage between the results of the Time-aware Retriever, the Fact Retriever, and ground truth in different question types in Figure 4. Across various question types, our Time-Aware Retriever answers most questions correctly that the Fact Retriever does, while also handling a significantly larger set of complex questions (second row). This indicates that after fine-tuning, the Time-Aware Retriever can retrieve more temporal information, while the Fact Retriever struggles to capture temporal information in the facts, leading to low answer coverage in the retrieved results.

5.5 Case Study

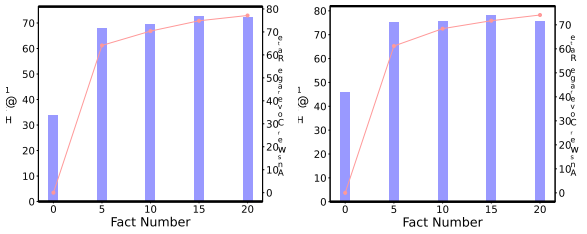
We provide examples for each question type in the MULTITQ dataset of ChatGPT *w/ TimeR*^{4}, LLaMA2 *w/ TimeR*^{4}, and TimeR^4 in Table 7 with the same input.

It can be observed in the results that without fine-tuned, LLMs tend to randomly generate a list of irrelevant answers, but in comparison, ChatGPT performs better than LLaMA2. LLaMA2 is more prone to generating irrelevant responses without fine-tuning. For instance, when asked *"After Okada Katsuya, who wishes to visit Cambodia first?"*, LLaMA2 directly outputs the entity mentioned in the question: '*Okada Katsuya*'. TimeR^4, on the other hand, achieves significantly higher accuracy and only generates the relevant answers.

Furthermore, after fine-tuning, the output of TimeR^4 adheres more closely to standard conventions. For questions like *"in which month...,"* where the ground truth answer is *['2012-05']*, LLMs of-

Model	MULTITQ					Timequestions				
	Overall	Question Type		Answer Type		Overall	Question Type		Answer Type	
		Single	Multiple	Entity	Time		Explicit	Implicit	Temporal	Ordinal
TimeR[4]	**72.8**	**88.7**	33.5	**63.9**	**94.5**	**78.1**	**82.3**	**73.0**	**83.0**	**64.9**
LLaMA2	18.5	22.0	10.1	23.9	5.5	28.9	26.8	41.9	33.7	33.8
LLaMA2 *w/ finetuned*	33.9	38.4	22.7	45.0	7.8	45.8	44.4	46.0	51.9	37.8
LLaMA2 *w/ TimeR*[4]	39.1	44.2	26.6	37.0	44.2	59.3	57.4	51.5	73.4	41.0
ChatGPT	10.2	14.7	7.7	13.7	2	45.9	43.3	51.1	46.5	42.1
ChatGPT *w/ TimeR*[4]	41.4	58.5	**41.2**	56.1	57.1	64.8	66.0	52.9	77.6	45.5

表6：将TimeR[4]框架与不同大型语言模型(LLM)结合用于推理的效果。



(a) Results on MULTITQ. (b) Results on TimeQuestions.

图3：不同事实数量的 Hits@1 结果。

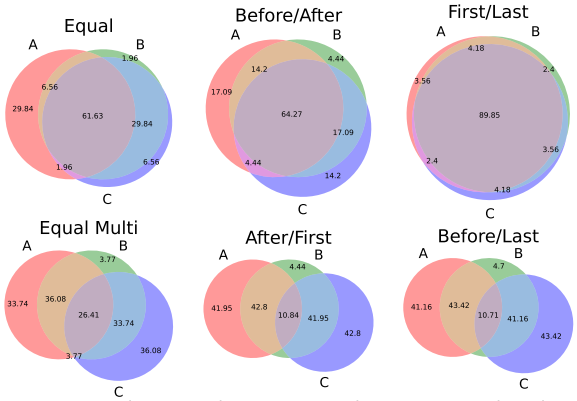


图4：MULTITQ 数据集中六种问题类型的时间感知检索、事实检索与真实答案覆盖重叠的维恩图。

5.4.2 检索到的事实数量

为了探索检索事实数量的变化对结果的影响，我们在图3中通过调整检索事实的数量 n 展示了两组数据集上的性能变化及相应的答案覆盖率。显而易见，模型在使用15条相关事实时性能达到峰值，这与我们策略中包含的数量相同。有趣的是，在 $n = 20$ 时性能略有下降，尽管使用20条检索事实获得了更大的答案覆盖率。这表明，过多的事实可能会通过引入噪声而阻碍性能，从而使大型语言模型更难区分相关将有用信息与无关信息区分开。另一方面，信息太少无法提供足够的时间知识，也会导致性能不佳。因此，$n = 15$ 是最佳选择。

5.4.3 检索的有效性

为了验证我们时间感知检索模块的有效性，我们在图4中调查了时间感知检索器、事实检索器以及不同问题类型的真实答案覆盖率之间的重叠情况。在各种问题类型中，我们的时间感知检索器能够正确回答大部分事实检索器能回答的问题，同时也能处理更多复杂问题（第二行）。这表明经过微调后，时间感知检索器能够检索到更多的时间信息，而事实检索器在捕捉事实中的时间信息方面存在困难，导致检索结果中的答案覆盖率较低。

5.5 案例研究

我们在表7中提供了MULTITQ数据集中每种问题类型的示例，这些示例分别来自ChatGPT *w/ TimeR*[4]、LLaMA2 *w/ TimeR*[4] 和 TimeR [4]，并使用相同的输入。

可以从结果中观察到，如果没有微调，LLM往往会随机生成一系列无关的答案，但相比之下，Chat-GPT 的表现优于 LLaMA2。LLaMA2在没有微调的情况下更容易生成无关的响应。例如，当被问到 *"After Okada Katsuya, who wishes to visit Cambodia first?"* 时，LLaMA2 直接输出问题中提到的实体：‘*Okada Katsuya*’。另一方面，TimeR[4] 的准确性显著更高，并且只生成相关的答案。

此外，在微调之后，TimeR[4] 的输出更接近标准惯例。对于像 *"in which month...,"* 这样地实际答案为 *['2012-05']* 的问题，LLM 会

Type	Question	Response		
		ChatGPT *w/ TimeR*^{4}	**LLaMA2 *w/ TimeR*^{4}**	**TimeR^4**
Equal	In 2005-01, who used light weapons to attack Thailand?	-Insurgent (Thailand) **-Citizen (Thailand)**	1. Insurgent (Thailand) 2. **Citizen (Thailand)** 3. Armed Opposition (Thailand)	**['Citizen (Thailand)', 'Armed Gang (Thailand)']**
Before/ After	Who investigated China after 22 July 2015?	1.Police (South Korea) 2. Mainland Affairs Council 3. Police (South Africa)	1. Japan 2. South Korea 3. France 4. China	**['Xi Jinping']**
First/ Last	In which month did Benny Gantz first visit China?	- **May** - August	**May**, August, or July	**['2012-05']**
Equal Multi	Who was the first to praise Iraq in 2015?	Iran	1. Iran 2. Iraq 3. el-Tayeb	**['Foreign Affairs (France)']**
After /First	After Okada Katsuya, who wish to visit Cambodia first?	- South Korea - Thailand - **Foreign Affairs (South Korea)**	1. Okada Katsuya 2. John Faulkner 3. Anupong Paochinda	**['Foreign Affairs(South Korea)']**
Before /Last	Who did Zimbabwe's Foreign Minister praise last before Kuwait?	- South Sudan - **Iran**	Guy Scott Mark Simmonds Tony Blair Faith Pansy Tlakula	**['Iran']**

Table 7: Comparison of responses to six different question types between our TimeR^4 and ChatGPT *w/ TimeR*^{4}, LLaMA2 *w/ TimeR*^{4}. The correct answers are highlighted in **bold font**.

ten generate *"May"*, which, although correct, is considered incorrect during evaluation. Exploring methods to construct a more reasonable evaluation or to ensure LLMs' outputs conform to standard conventions will be our future research. These results demonstrate the effectiveness of TimeR^4 in conducting temporal reasoning.

6 Conclusion

In this work, we address two key challenges that LLMs face when handling temporal questions and introduce a time-aware retrieve-rewrite-retrieve-rerank framework named TimeR^4.

To mitigate the issue of the temporal hallucination of LLMs, we utilize a *retrieve-rewrite* strategy to fetch relevant facts in FKS and integrate specific timestamps into the questions. Afterward, in order to retrieve facts that satisfy both time constraints and semantic similarity, we implement a *retrieve-rerank* strategy. We perform time-aware retrieval from the TKS and rerank them based on temporal constraints. Finally, we fine-tune open-source LLMs, leveraging the knowledge in TKGs to enhance the temporal reasoning capabilities in LLMs. Experiments on two datasets demonstrate that our framework significantly enhances the temporal reasoning abilities of LLMs.

Limitations

Although our approach achieves significant improvements, with relative gains of 47.8% and 22.5% on two TKGQA datasets, the performance of complex temporal fact retrieval can still be further enhanced. How to retrieve more precise and effective temporal information is a question worthy of exploration.

Additionally, controlling the answer format during the generation process of LLMs without fine-tuning is difficult, as discussed in Section 5.5. Thus, standardizing the answer formats of LLMs or developing a more reasonable evaluation method is another important future task.

7 Acknowledgements

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Yejin Bang, Samuel Cahyawijaya, Nayeon Lee, Wenliang Dai, Dan Su, Bryan Wilie, Holy Lovenia, Ziwei Ji, Tiezheng Yu, Willy Chung, et al. 2023. A multitask, multilingual, multimodal evaluation of chatgpt

Type	Question	Response		
		ChatGPT *w/ TimeR*^{4}	**LLaMA2 *w/ TimeR*^{4}**	**TimeR^4**
Equal	In 2005-01, who used light weapons to attack Thailand?	-Insurgent (Thailand) **-Citizen (Thailand)**	1. Insurgent (Thailand) 2. **Citizen (Thailand)** 3. Armed Opposition (Thailand)	**['Citizen (Thailand)', 'Armed Gang (Thailand)']**
Before/ After	Who investigated China after 22 July 2015?	1.Police (South Korea) 2. Mainland Affairs Council 3. Police (South Africa)	1. Japan 2. South Korea 3. France 4. China	**['Xi Jinping']**
First/ Last	In which month did Benny Gantz first visit China?	- **May** - August	**May**, August, or July	**['2012-05']**
Equal Multi	Who was the first to praise Iraq in 2015?	Iran	1. Iran 2. Iraq 3. el-Tayeb	**['Foreign Affairs (France)']**
After /First	After Okada Katsuya, who wish to visit Cambodia first?	- South Korea - Thailand - **Foreign Affairs (South Korea)**	1. Okada Katsuya 2. John Faulkner 3. Anupong Paochinda	**['Foreign Affairs(South Korea)']**
Before /Last	Who did Zimbabwe's Foreign Minister praise last before Kuwait?	- South Sudan - **Iran**	Guy Scott Mark Simmonds Tony Blair Faith Pansy Tlakula	**['Iran']**

表7：我们 TimeR^4 与 ChatGPT *w/ TimeR*^{4}、LLaMA2 *w/ TimeR*^{4} 对六种不同问题类型的回答比较。正确答案以粗体显示。

十个生成 *"May"*，虽然正确，但在评估中被认为是不正确的。探索构建更合理评估的方法或确保大型语言模型的输出符合标准规范将是我们未来的研究方向。这些结果展示了 TimeR^4 在进行时间推理方面的有效性。

6 结论

在这项工作中，我们解决了大型语言模型在处理时间性问题时面临的两个关键挑战，并提出了一个名为 TimeR^4 的时间感知检索-重写-重检索-重排序框架。

为了缓解大语言模型（LLMs）的时间幻觉问题，我们利用 *retrieve-rewrite* 策略从 FKS 中获取相关事实，并将具体时间戳整合到问题中。随后，为了检索同时满足时间约束和语义相似性的事实，我们实施了 *retrieve- rerank* 策略。我们从 TKS 中进行时间感知检索，并根据时间约束重新排序。最后，我们对开源大语言模型进行微调，利用 TKGs 中的知识提升 LLMs 的时间推理能力。在两个数据集上的实验表明，我们的框架显著增强了 LLMs 的时间推理能力。

限制

虽然我们的方法取得了显著的改进，在两个 TKGQA 数据集上分别获得了 47.8% 和 22.5% 的相对增益，但性能复杂时间事实检索仍然可以进一步增强。如何检索更精确和有效的时间信息是一个值得探讨的问题。

此外，如第5.5节所述，在不进行微调的情况下控制大型语言模型（LLMs）生成过程中的答案格式是困难的。因此，标准化大型语言模型的答案格式或开发更合理的评估方法是另一个重要的未来任务。

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A Prompt Template

The prompts for rewriting can be found in Figure 5. The template used for instruction tuning and reasoning is shown in Figure 6.

Rewriting Prompt Template

Replace the temporal fact in questions with explicit timestamps from the provided facts or your knowledge without any explanation. If you are not sure about the answer, return the original questions.

For instance, from the fact:
"[Juan Carlos I, Praise or endorse, Vietnam, 2006-02-22]",
We can modify the question:
"After Vietnam, who was the first to praise Juan Carlos I?"
to "After 2006-02-22, who was the first to praise Juan Carlos I?"

Here is your turn:
Facts: `<fact>`
Question: `<question>`

Figure 5: Rewriting prompt template.

Reasoning Prompt Template

Based on the historical facts, please answer the given question. Please keep the answer as simple as possible and return all the possible answers as a list.
Historical facts: `<fact>`
Question: `<question>`

Figure 6: Reasoning prompt template.

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Rewriting Prompt Template

Replace the temporal fact in questions with explicit timestamps from the provided facts or your knowledge without any explanation. If you are not sure about the answer, return the original questions.

For instance, from the fact:
“[Juan Carlos I, Praise or endorse, Vietnam, 2006-02-22]”,
We can modify the question:
“After Vietnam, who was the first to praise Juan Carlos I?”
to “After 2006-02-22, who was the first to praise Juan Carlos I?”

Here is your turn:
Facts: 〈fact〉
Question: 〈question〉

图5：重写提示模板。

一个提示模板

用于改写的提示可以在图5中找到。用于指令调优和推理的模板如图6所示。

推理提示模板

根据历史事实，请回答所给问题。请尽量保持答案简洁，并将所有可能的答案以列表形式返回。历史事实：〈fact〉问题：〈question〉

Figure 6: Reasoning prompt template.